17-626: Requirements for Information Systems
Assignment 3: Spec-Driven Development

Software engineering (SE) practice is transforming to use SE agents that take requirements as inputs, called "specs," and generate code as output. This permits developers to focus more on writing the right requirements and on quality assurance than on writing code. In this assignment, students will practice **spec-driven development** using GitHub Spec Kit and an AI-assisted development tool. Students will use stakeholder interviews and goal models to identify and prioritize requirements, develop specifications, build prototypes, and evaluate how well the resulting implementation supports stakeholder goals.

The objective is not simply to build a working application, but to understand how human judgment contributes to requirements selection, conflict resolution, traceability, and specification refinement, when AI can generate most of the implementation.

Required Readings

- GitHub Spec Kit QuickStart[1] and Command Reference[2]

Learning Objectives

- Guide SE agents in translating stakeholder requirements into code
- Trace and evaluate whether code satisfies requirements and discover new requirements through implementation
- Identify and distinguish complementary roles for human developers and SE agents in software design

Assignment Deliverables

You are a software engineer assigned to a team at ABC Design Services, who is responsible for elicitation, requirements modeling and analysis, sprint planning, and requirements verification. Your team has recently completed goal modeling from stakeholder interviews. Your task is to review the goal models and plan two prototypes, after which you will supervise an SE agent in the implementation of the prototypes to ensure the software satisfies the associated requirements. Your employer is very optimistic that SE agents will replace human developers, so your task is to carefully monitor your agent(s) and ensure they are doing good work, while also documenting your role throughout the process.

Your job is to be faithful to stakeholder needs and expectations, anticipate additional needs and risks, and validate that "additional needs" are necessary or desired.

Because your company does not use a formal process (e.g., Scrum), you have wide latitude to decide what portion of the requirements that you have collected will be included in your first deliverable. You should choose goals that demonstrate a *core feature*, a *user-facing aspect* or an *aspect with high uncertainty* and clearly justify your choice. This selection will be used in a mock sprint to develop this feature or aspect.

---

[1] https://github.github.io/spec-kit/quickstart.html
[2] https://github.github.io/spec-kit/reference/agentic-sdd.html

Step 1: Review the goal models that your team members created in Assignment #2. Based on the three priority criteria (core, user-facing, uncertain), choose a subset of the goals from your stakeholders to become the system boundary for your prototypes. Merge and reconcile any differences between your team members' goal models: one team member may have identified more detail than another on a particular aspect of the system, or this detail may be distributed across multiple team members. Combine your findings to realize the best model for this assignment. Justify your decision by considering other priorities that you decided were less important and *include those in your learning log, below*.

Step 2: Review the relationships between the selected requirements and stakeholder goals. Identify any meaningful overlap, dependency, or conflic**t** involving your selected requirements. For example, two stakeholders may request different behaviors for the same feature, or two requirements may support the same goal but propose different approaches. Use your goal model to explain the relationship and justify your decision. Where requirements overlap, determine whether they should be combined, implemented separately, or prioritized differently. Where goals conflict, explain the trade-off and how you decided to proceed. You may prioritize one goal, develop a compromise, or document an unresolved conflict that requires additional stakeholder clarification. **Do not simply remove conflicting goals or requirements from the model.** Preserve the relationship and document the reasoning behind your decision. Update your goal model to reflect your sprint scope and any decisions or assumptions made.

Step 3: Create your development environment by creating a GitHub private repository GitHub Spec Kit and an AI-assisted development tool, translate your selected requirements into a "spec". Your initial spec should include:

- The selected stakeholder goals and corresponding requirements.
- Acceptance scenarios defining how the requirements can be evaluated.
- Relevant assumptions, dependencies, and edge cases.
- Decisions made regarding overlapping or conflicting requirements.

Use Spec Kit to generate an implementation plan and development tasks, then build a functional prototype based on the specification. **You should use mock data and simplified interactions where appropriate**. The prototype should only demonstrate the selected requirements rather than attempt to implement all elicited goals. Preserve your initial specification, goal model, Spec Kit artifacts, and prototype source code as **Version 1 (V1)** in GitHub. Deploy Prototype V1 and retain its URL.

Step 4: Interact with Prototype V1 and evaluate whether it satisfies the requirements you selected for the sprint. Review and assure you have traceability between the interview transcripts, goal model, specification, and implementation. For each selected requirement, identify:

- The supporting stakeholder interview evidence.
- The corresponding goal in your goal model.
- The requirement identifier in your specification.
- The relevant code component, function, or implemented behavior.
- Whether the requirement is fully satisfied, partially satisfied, or not satisfied.

Document these relationships in a **Requirements Traceability Matrix**, like the one below:

Requirement[3]	Interview Evidence[4]	Goal Model Reference[5]	Code / Implementation	Status
REQ 01	Transcript Reference	Goal ID	Component/Function	Fully satisfied
REQ 02	Transcript Reference	Goal ID	Component/Function	Partially satisfied
REQ 03	Transcript Reference	Goal ID	Not implemented	Not satisfied

Identify any **meaningful observations** from your evaluation. For each observation, distinguish between:

- **Implementation defect:** The prototype does not satisfy a requirement that was already clearly specified.
- **Requirements issue:** The specification is missing information, or contains ambiguous, incorrect, or insufficient information to satisfy the stakeholder's goal.

As you evaluate the prototype, consider and report your answers in your learning log, below:

- Describe what you discovered.
- Identify whether it is a requirements issue or an implementation defect.
- Explain what needs to change.

Explain what each observation reveals and whether a specification change, implementation correction, or further stakeholder clarification is needed.

Step 5: Use the findings from your evaluation to improve the specification. Revisit your original requirements decisions, including any conflicts or overlaps identified earlier. Update the goal model where your understanding of goal relationships or satisfaction has changed. Preserve unresolved stakeholder conflicts and document any assumptions or trade-offs. Revise the specification before changing the intended implementation behavior. Use the revised specification to update the relevant planning artifacts and prototype. Preserve the revised specification, goal model, and implementation as **Version 2 (V2)** in the same GitHub repository. Deploy Prototype V2 separately so that both versions can be accessed and compared. Your revised version should demonstrate at least one meaningful requirements-level improvement informed by evaluating Prototype V1.

Step 6: Prepare a single document named learning-log.md describing your decisions, observations, and learning throughout the sprint.

---

[3] From Assignment 1
[4] From Assignment 1
[5] From Assignment 2

The document should contain the following sections.

1. Sprint Planning and Requirements Selection

Briefly describe the requirements and the goals you selected for the sprint, and why you prioritized them. Identify the overlapping, dependent, or conflicting requirements you considered. Briefly explain how your goal model helped you understand these relationships, what decisions you made, and why.

2. Prototype Evaluation

Discuss your significant observations from Prototype V1. Explain whether the issues arose from the specification or implementation and what they revealed about the stakeholder goals.

3. Revisions and Learning

Describe the significant changes between Versions 1 and 2, including changes to the goal model, specification, and prototype. For each significant change, explain:

- What you initially intended.
- What you discovered through evaluation.
- What you changed and why.
- How the revised implementation better addresses the requirement or stakeholder goal.

Conclude with a brief reflection addressing:

- Which decisions required your judgment rather than relying solely on AI?
- How did prototyping change your understanding of stakeholder needs?
- How did spec-driven development help maintain consistency between the requirements and implementation?
- **What evidence from your work would you present to management to demonstrate the value you contributed beyond AI-generated specifications and code?**

The learning log should emphasize your reasoning and decisions rather than simply listing the commands or prompts used during development.

Assignment Submission

Assignments are due at the date and time shown in the course website. Submit electronically at the course website. Submit **one GitHub repository URL** through the course website. Your repository must contain the complete source code, goal models, Spec Kit artifacts, requirements traceability matrix, and learning document. Maintain both versions in the **same repository**, preferably using separate v1 and v2 branches/tags. Version 2 should build upon Version 1 so that the development history is preserved. Your repository's default branch must contain a clearly organized README.md with direct links to the following five components:

Deliverable	Description
1. GitHub Version 1	Direct link to the v1 branch or commit containing the initial goal model, specification, Spec Kit artifacts, and prototype source code.
2. GitHub Version 2	Direct link to the v2 branch or commit containing the revised goal model, specification, Spec Kit artifacts, and prototype source code.
3. Prototype V1 URL	A working deployment of the initial prototype.
4. Prototype V2 URL	A working deployment of the revised prototype.
5. Requirements Analysis and Learning Log	Direct link to the document containing requirements selection, goal relationships, traceability, evaluation findings, revisions, and reflection.

Recommended Repository Structure

```
None
repository/
├── frontend/              # Application source code
├── specs/                 # Spec Kit artifacts
├── docs/
│   ├── goal-model.md      # Goal model and decisions
│   ├── traceability.md    # Requirements Traceability Matrix
│   └── learning-log.md    # Requirements analysis and reflection
└── README.md              # Links to both versions and deployments
```

Each version must preserve its corresponding specification, goal model, and implementation artifacts. The README.md should make it easy to navigate between versions and understand what changed. Both deployed prototypes must remain functional and accessible at the time of evaluation. **Only the GitHub repository URL needs to be submitted.** All five deliverables must be accessible through its README.md.

AI Assistance

You may use AI assistance to complete this assignment, except for the authorship of the learning log. All examples and answers to the questions in Step 6 must be completed without the use of AI assistance. Any indication of using AI assistance on completing the learning log will be handled in accordance with the AI Assistance Policy in the course syllabus.

Evaluation Criteria

Your work will be evaluated on your ability to:

1. **Select and justify requirements:** Identify a coherent sprint scope grounded in stakeholder evidence and goal models.
2. **Analyze goal relationships:** Recognize overlapping, dependent, and conflicting requirements and justify the decisions made.

3. **Apply spec-driven development:** Produce coherent specifications, planning artifacts, and working prototypes.
4. **Establish traceability:** Connect stakeholder interview evidence to goals, requirements, and implemented functionality.
5. **Evaluate and revise:** Identify meaningful requirements issues, improve the specification and goal model, and demonstrate the resulting changes in Prototype V2.
6. **Demonstrate human judgment:** Clearly explain how your reasoning, decisions, and evaluation contributed to the outcome beyond AI-generated artifacts.

The emphasis is on requirements engineering, traceability, and evidence-based decision-making, rather than the visual polish or technical complexity of the prototype. **Remember:** Your goal is not simply to demonstrate that you can build software using AI coding assistance. Your goal is to demonstrate how and why understanding stakeholder needs, making informed requirements decisions, and evaluating what was built remain essential responsibilities of a software engineer.